CHAPTER 1

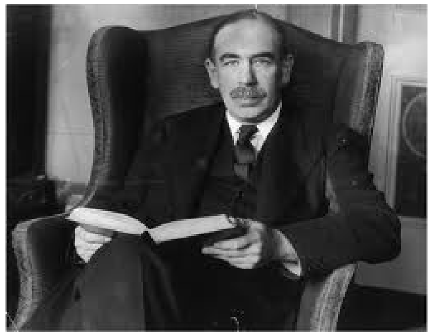
Population/Process and Sample

1. What is Statistics?

Ronald Aylmer Fisher (who is often referred to as the "father of statistics") considered statistics a branch of applied mathematics.

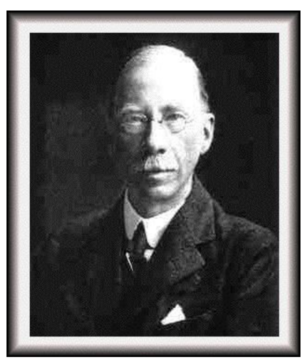
Ronald Aylmer Fisher

As stated in Fisher (1958, pp 1) "the science of statistics is essentially a branch of Applied Mathematics, and may be regarded as mathematics applied to observational data." He goes on to state "as other mathematical studies, the same formula is equally relevant to widely different groups of subject-matter. Consequently the unity of the different applications had usually been overlooked, the more naturally because the development of the underlying mathematical theory had been much neglected." It is not clear what he meant by "applied" mathematics (but its not clear what most people mean when they place "applied" in front of mathematics). None the less, "statistical thinking" would then be just a special case of "mathematical thinking." A variety of researchers and practitioners make use of statistical methods to analyze their data. These applications, include among others, environmental, pharmaceutical, legal, biological, political, and industrial data. The techniques used to analyze these data range from simple graphical representations of data to quite involved formal statistical methods. Fisher (1958) further regarded statistics as "(i) the study of populations, (ii) as the study of variation," and "(iii) as the study of methods of the reduction of data."



John Maynard Keynes

John Maynard Keynes (1920) wrote in his book *A Treatise on Probability* that "1. The Theory of Statistics, as it is now understood,[1] can be divided in two parts which are for many purposes better kept distinct. The first function of the theory is purely *descriptive*. It devises numerical and diagrammatic methods by which certain salient characteristics of large groups of phenomena can be briefly described; and it provides formulae by the aid of which we can measure or summarise the variations in some particular character which we have observed over a long series of events or instances. The second function of the theory is *inductive*. It seeks to extend its description of certain characteristics of observations events to the corresponding characteristics of other events which have not been observed. This part of the subject may be called the Theory of Statistical Inference; and it is this which is closely bound up with the theory of probability. 2. The union of these two distinct theories in a single science is natural." Keynes' reference is to George Udny Yule's book titled *Introduction to the Theory of Statistics*.



George Udny Yule

2. Population and Sample

We will take the view that a researcher is interested in studying a process. All possible individuals or items that this process has or could have produced will be referred to as the **population** of interest. Under this definition, the population associated with a process is an uncountable collection. The individuals the process has produced constitutes a representative **sample** of what the process can produce.

Often the representative sample is referred to as the "population." Further, it is usually of interest to study the (joint) **distribution** of one (or more) **variable**(s) and/or **measurement**(s) to be taken on individual members of the population. The measurement(s) taken on an individual is sometimes referred to as an observation on the individual. The collection of observations on the individuals of a population are also referred to as the population (see Fisher (1958)). For example, a manufacturer of adult men clothing is interested in studying the process that produces adult men. Presently, there is available a very large (but finite) sample of adult men that the process has produced. Clearly, the manufacturer is not only interested in this sample but the men the process could have produced which will include those it will produce. Two measurements of interest are inseam and waist length, in inches, of each adult man.

As a second example, a pollster will be interested in the proportion of individuals who will say they will vote for a given presidential candidate one month before the actual election. At that point in time there will be a finite number of voters. So how does one view this collection as a representative sample of what some process has produced and not the population of interest? What if the election to be studied by the pollster is to occur four years from now? The collection of voters that will exist one month before the election date has yet to be (completely) generated. Consequently, this collection of voters can only be viewed as a representative sample of what the process can generate one month before the election as well as at any time. It then follows that a **census** occurs when the measurement(s) of interest is taken on each individual in the representative sample.

Fisher (1958, pp 33) states "when a large number of individuals are measured in respect of physical dimensions, weight, color, density, etc., it is possible to describe with some accuracy the *population* of which our experience may be regarded as a sample. By this means it may be possible to distinguish it from other populations differing in their genetic origin, or in environmental circumstances." Further, Fisher (1958, pp. 41) states the following about population and frequency distribution. "The idea of an infinite **population** distributed in a **frequency distribution** in respect of one or more characters is fundamental to all statistical work. From a limited experience, for example, of individuals of a species, or of the weather of a locality, we may obtain some idea of the infinite hypothetical population from which our sample is drawn, and so of the probable nature of future samples to which our conclusions are to be applied." He alludes to a process (genetic origin or environmental circumstances), describes a population as an infinite collection, and does not make a distinction between the population as a collection of individuals or as the collection of the measurements on these individuals.

William Edwards Deming (October 14, 1900 – December 20, 1993) is a well known statistician. Sampling techniques that he developed are still used today by the United States Department of the Census and the Bureau of Labor Statistics. He is best known and recognized by the Japanese for his post World War II work in helping the Japanese capture world markets in several key industries such as automotive, steel, and electronics. Their success was due to producing high quality products.

Deming (1953) stated the following. "Statistical data are supposedly collected to provide a rational basis for action. The action may call for the enumerative interpretation of the data, or it may call for the analytic interpretation. The aim here

FIGURE 1

is to exhibit some of the consequences of failing to distinguish between the enumerative and the analytic uses of data. This distinction is necessary in the statement of the aims of a survey, census, or experiment, in order that the plans for the collection of the data and for the tabulations may most economically meet the needs of the consumer, and it is equally important in the interpretation of data. Thus, to draw on a result from a later paragraph, information obtained in a complete census concerning every person in an area (e.g., on occupation, income, or education) still possesses for analytic purposes a sampling error that is actually about a quarter as great as the sampling error of a 6 per cent sample. The consequences are far-reaching. In using a census-table for analytic purposes, even though the figures come from a perfect complete count, it is therefore necessary to bear in mind that small numbers in a cell are unreliable in the sense that they have a standard error, just as if they had arisen in sampling, as indeed they did. Moreover, in the planning of a complete census, it is therefore imperative to use sampling for every bit of information that is not necessary as an aid to complete coverage, or required to give detail for small areas (such as the block statistics). Name, relationship to the head, age, sex, marital status, color are probably all necessary for the sake of completeness of coverage. These things, plus a few questions on rent, tenure, year built, will provide the information required for the block statistics." He points out that even a census is a sample. Recall that a census occurs when the measurement(s) of interest is taken on each individual in the representative sample. The implication here is that the population is a collection of individuals that contains the individuals that have been generated by a process.

Montgomery (2001) gives an example of a process packaging frozen orange juice concentrate in a 6-oz cardboard container. One measurement of interest is whether the container leaks or not. A number that completely characterizes the distribution of this measurement is the rate, p, at which the process is producing containers that leak. Another equivalent view of the meaning of p is that p is the probability that a container will leak. Since the number p characterizes the population of interest it is generally referred to as a **parameter**. If N of these containers of frozen orange juice concentrate produced by this process are to be purchased by a store, the N items only constitute a representative sample from the process. There is no reason for the store to view this collection of N items as a population. The store may be interested in the number or proportion of these N containers that leak, but these values will depend on the sample received. Such numbers that characterize a sample are referred to as **statistic**s.

Douglas C. Montgomery

It is useful to classify variables/measurements of interest. Virtually all of the variables we will study can be classified as either discrete or continuous. A **discrete** variable is a variable in which its set of possible values is a countable set. This collection of possible values of a discrete variable are either a finite or countably infinite set. Measurements such as length, weight, time, area, volume, and rates are variables in which the set of possible outcomes is a continuum of the real numbers, typically, some interval of the real numbers These are examples of **continuous** variables. In the orange juice concentrate example, the measurement of interest has only two possible outcome – leaks or does not leak. The example about adult men both measurements of interest are continuous variables.

Often the process/population of interest is studied by examining the (joint) distribution of the measurement(s) of interest. A useful way to view the distribution of a continuous variable is with a **density curve** (function). For many continuous variable X of interest, the density (function) curve describing its distribution is simply a (function) curve $y = f(x)$ such that the curve is on or above the measurement axis and the total area under the curve is one. Further, area under the curve and over an interval is the proportion of the population having their measurement X falling in this interval. That is, the more dense the measurements from the population in some interval the larger the area under the density curve and above this interval.

It is of importance to keep in mind that the researcher is interested in the process and the associated population. The population is a infinite collection and hence the researcher could never view the entire population. Since one cannot view the whole population, then only partial information about the process/population can be obtained by looking at a sub-collection of what the process can produce. This sub-collection is referred to as a **sample**. Ideally, we would like for this sample to be representative of the population. What the process has produced is a **representative sample** of what the process can produce. Often this sample is so large that it would be impossible for the researcher to view this sample. One solution is to take a representative sample from this representative sample. Unfortunately, there is no sampling method that guarantees a representative sample. The best for which one can hope is a method that will "most likely" produce a representative sample. Sampling methods will be discussed further in a later chapter. In this chapter and the next, we will assume when we have a sample the method for producing this sample selected the sample in such a way that there is a good chance the sample is representative of the population.

EXERCISE 1. *A coin is to be flipped. A measurement X is taken on the coin with X having the value 1 if the face-up side of the coin is a "head" and a value of 0 if the face-up side of the coin is a "tail." The process is the flipping of the coin. The population is the collection of possible flips of the coin. This is an uncountable collection of possibilities. For the American penny, it is common to assume that the outcomes of a "head" ($X = 1$) and a "tail" ($X = 0$) are equally likely (fifty-fifty). Assume this model. Suppose that the coin has been flipped but the outcome of "head" or "tail" has not been revealed. Does the knowledge that the coin has been flipped provide information that would cause one to now assume a different model than the fifty-fifty model?*

EXERCISE 2. *There is a process that produces rough sawed two-by-fours that are ten feet long. This is a piece of lumber that is two inches by four inches and ten feet long. The population of interest is all the two-by-fours the sawing process has produced or could have produced. Let X be the actual width of the board which has a target width of two inches. A sample of size $n = 2$ two-by-fours is to be selected from the output of the process. Let X_1 and X_2 represent the widths of these boards. Define the statistics Y_1 and Y_2 by*

$$Y_1 = X_1 + X_2 \text{ and } Y_2 = X_1 \times X_2.$$

The boards have been selected and their widths are $x_1 = 2$ inches and $x_2 = 2$ inches. Are the observed values y_1 and y_2 of the statistics Y_1 and Y_2 equal? Why or why not?

EXERCISE 3. *A penny is to be placed on a flat surface, spun, and its resting side that is up is to be observed. This is a process of generating "heads" and "tails." The collection of all possible ways of spinning this penny is an uncountable collection. Let p be the rate at which the process is generating spins that result in a resting side up of "heads." Define the measurement X to have the value of 1 if the spin of the penny results in the coin resting heads-up and $X = 0$ if tails-up. The coin is now spun two hundred times and it is observed that the number of times the coin was "heads" in its resting position was 83. Thus, the proportion of the 200 spins that resulted in "heads" is $83/200 = 0.415$. Is $p = 0.415$?*

EXERCISE 4. *Suppose a finite population exists. Can a measurement X on the individuals of this population be a continuous variable? No, since there is a finite number of individuals in the population, then there can only be a finite number of possible values for the measurement X. Hence, it is a discrete measurement.*

EXAMPLE 1. *Students are reluctant to report cheating by other students. A student project put this question to a simple random sample (SRS) of $n = 172$*

students attending the University of Mea Colpas (UMC): "You witness two students cheating on a quiz. Do you go to the professor?" The number Y_n *who answerd "Yes" was observed to be* $y_n = 19$. *The proportion of the* $n = 172$ *students who answered "Yes" is*

$$\widehat{p}_n = \frac{y_n}{n} = \frac{19}{172} \approx 0.1104651163.$$

Clearly, the numbers Y_n *and* $\widehat{p}_n$ *are statistics. A process generated the* N *students attending UMC during the term in which the* $n = 172$ *were asked the question. Among these* N *students, let* Y_N *be the number who would answer "Yes" to the question if asked. Then* $\widehat{p}_N = Y_N/N$ *is the proportion of this collection who would answer "Yes." The collection of* N *students is representative of what the process can generate. This collection is the representative sample. Note that the numbers* Y_N *and* $\widehat{p}_N$ *are not observed. These numbers are statistics. We could use the statistic* $\widehat{p}_n = 19/172$ *to predict the value of* $\widehat{p}_N$ *and the value* $N\widehat{p}_n$ *to predict the value of* Y_N *if either of these numbers are of interest. If one is interested in estimating the rate* p *at which the process is generating students into this university who would answer "Yes" to the question (population proportion), the value* $\widehat{p}_n = 19/172$ *could be used.*

EXAMPLE 2. *A process has generated a collection of health adults. The population of interest is the collection of healthy adults the process has or could have generated. At three hour time intervals, the body temperatures* $X_1, X_2, \ldots, X_8$ *of each adult are to be measured over a 24-hour period and the average*

$$Y = \frac{X_1 + X_2 + \ldots + X_8}{8}$$

recorded. The measurement Y *(daily average body temperature) on each health adult is a continuous variable. Hence, there is a density curve that describes how* Y *is distributed over the population of healthy adults. It is generally assumed that average body temperature* μ, *in degrees Fahrenheit, is* 98.6° F. *This is the mean of the distribution of body temperature measurements* X *over all healthy adults. The assumption that* $\mu = 98.6^{\circ}$ F *is a model. As we will see later, the average* μ_Y *over all healthy adults is also* 98.6° F.

EXAMPLE 3. *Moore (2010) states "A density curve is often a good description of the overall pattern of a distribution. Outliers, which are deviations from the overall pattern, are not described by the curve. Of course, no set of real data is exactly described by a density curve. The curve is an idealized description that is easy to use and accurate enough for practical use." What does the author mean by a "set of real data?"*

EXAMPLE 4. *(continuation) Why are outliers not described by the the density curve?*

EXAMPLE 5. *Moore (2010) states that "a density curve is a curve that (i) is always on or above the horizontal axis, and (ii) has area exactly* 1 *underneath it. A density curve describes the overall pattern of a distribution. The area under the curve and above any range of values is the proportion of all observations that fall in that range." Consider any point with coordinates* (x, y) *on a density curve. What does the variable* x *represent?*

EXAMPLE 6. *(continuation) Can the value of the variable* y *be negative? Why or why not?*

3. Other Views of Population and Sample

Moore, Notz, and Fligner (2013) writes in Chapter 17 on page 438 that "There is another condition that applies to all of the inference methods in this book: *the population must be much larger than the sample, say at least 20 times as large.*[1] All of our examples and exercises satisfy this condition. Practical settings in which the sample is a large part of the population are rather special, and we will not discuss them." ([1] Chapter 18, Note 1: Note 2 for Chapter 11 explains the reason for this condition in the case of inference about a population mean. Chapter 11, Note 2: Strictly speaking, the formula $\sigma/\sqrt{n}$ for the standard deviation of $\overline{x}$ assumes that we draw an SRS of size n from an *infinite* population. If the population has finite size N, this standard deviation is multiplied by $\sqrt{1-(n-1)/(N-1)}$. This "finite population correction" approaches 1 as N increases. When the population is at least 20 times as large as the sample, the correction factor is between about 0.97 and 1. It is reasonable to use the simpler form $\sigma/\sqrt{n}$ in these settings." This implies that he believes that all the populations in which he will give examples are finite collections. Moore (2010) makes an interesting statement on page 597 "Because the conclusions of inference always concern some *population*, the conditions describe the population and how the data are produced from it." On page 248 of Moore and McCabe (2003), the authors state "The entire group of individuals that we want information about is called the **population**." Moore (2010) in Example 17.2 on page 447 discusses Example 14.3 found on page 366. He states in this example that "Biologists studying the healing of skin wounds measured the rate at which new cells closed a razor cut made in the skin of an anesthetized newt."

David S. Moore George P. McCabe

At the time of this study, there were a finite but unknown number N of newts in existance. Let's represent them by

$$Newt_1, Newt_2, \ldots, Newt_N.$$

This is a finite collection. Is this the entire group of individuals about which we want information? If so, according to Moore and McCabe (2003), this collection is then the population. According to Fisher (1958), this would be what we have referred to as the representative sample if they had all be anesthetized and razor cut. However, none of these N newts have been anesthetized and razor cut by the researcher. Consequently, there are no "actual" members in the population just before the study begins. The only "actual" newts who will have been anesthetized and razor cut by the researcher will be those in the researcher's sample.Today, there is a finite number N^* newts in existence which could be represented by

$$Newt_1^*, Newt_2^*, \ldots, Newt_{N^*}^*.$$

While some of these may be among the previously mentioned group of N newts, it seems reasonable to assume some new ones have been added and some have died. Is this now the population of interest if each one had been anesthetized and razor cut? If so, what was learned about, say for example, the average rate of healing of the previous collection of N should not apply to the present group of N^*? Unless, we view both collections as representative samples generated by the process that generates newts and our population of interest are all the newts the process has or could have geneated who will be anesthetized and razor cut.

Moore (2008) on page 620 states "The two-sample t procedures of Chapter 19 compare the means of two populations or the mean responses to two treatments in an experiment. Of course, studies don't always compare just two groups. We need a method for comparing any number of means." Here he is referring to population means." Exercise 25.31 on page 650 of Moore (2008) states "How do nematodes (microscopic worms) affect plant growth? A botanist prepares 16 identical planting pots and then introduces different numbers of nematodes into the pots. He transplants a tomato seedling in each pot. Here are data on the increase in height of the seedling (in centimeters) 16 days after planting:

$$\begin{bmatrix} \text{Nematodes} & & \text{Seedling growth} & & \\ 0 & 10.8 & 9.1 & 13.5 & 9.2 \\ 1,000 & 11.1 & 11.1 & 8.2 & 11.3 \\ 5,000 & 5.4 & 4.6 & 7.4 & 5.0 \\ 10,000 & 5.8 & 5.3 & 3.2 & 7.5 \end{bmatrix}."$$

There are four populations of interest in this example. An individual is a planting pot containing i nematodes and a transplanted tomato seedling (population i) grown for 16 days, where $i = 0, 1000, 5000, 10000$. Note that the researcher is part of the process that generates the individuals in these populations. For each population, the number of actual individuals is $n = N = 4$. The representive sample is the researcher's sample as is often the case in a designed experiment. The mean

$$\overline{x}_{1000} = \frac{11.1 + 11.1 + 8.2 + 11.3}{4} = 10.425$$

is the mean of the researcher's sample. It is not the population mean μ_{1000}. The population $i = 1000$ is the collection of all individuals the process has (actual) or could have (conceptual) generated. The conceptual individuals are an uncountable collection. In this example, the researcher is interested in knowning μ_{1000} but must settle for an estimate $\overline{x}_{1000} = 10.425$.

Kempthorne and Folks (1971) state on page 11 that "The subject of statistics is aimed largely at the understanding of populations ... ." They suggest on page 44 that there exist finite populations. On the other hand on the next page, page 45 of their book, they write "The second portion of the book deals with the general problem that most of the data sets which we meet are not usefully regarded as comprising complete populations, but should be regarded as portions, obtained by a process called random sampling, of infinite populations. The activity is called inferential statistics, and within it are many different complementary and competing aspects." The authors discuss on page 158 under the topic of sampling distributions "a population of N individuals, for example, the undergraduate body of college X consisting of 1376 students." On page 160, they return to infinite populations and state "We now describe briefly the concept of a random sample from mathematical infinite populations such as the binomial and the normal." But on page 455, they state "In most practical data situations we will have available a definite amount of data, and our basis for interpreting the data will be that they have come from conceptual populations by some sort of random sampling." At the end of this page, they go on to write "Before taking up simple cases, it should be mentioned that there has been extensive development of elementary ideas of random sampling from finite structured populations. We refer the reader to Tukey (1950, 1956), Kempthorne (1952), Wilk and Kempthorne (1955, 1956, 1957), Cornfield and Tukey (1956), Kempthorne, et al. (1961), Zyskind (1962, 1963), Zysking et al. (1964)."

Oscar Kempthorne

Many authors seem to believe researchers are only interested in making inferences about finite collection of individuals. But a closer look at their writings suggest that they are using estimation methods to make an inference about the finite representative sample. What they discount is the uncertainty that is associated with the representative sample. They view, for example, the mean of the representative sample not as a sample mean but as a population mean. It may be of interest to know this sample mean. But any method that is uses the mean of the researcher sample to predict the mean of the representative sample must take into consideration the variability found in the mean of the representative sample.

EXAMPLE 7. *In the Problems for Chapter 2 of Kempthorne and Folks (1971) they state in Problem 1 "Fifty 8-ohm resistors were handpicked from standard production for use in a special apparatus being assembled by a small electronic firm.*

Each resistor is individually tested and the resistances for this population of resistors are as follows:

$$\begin{bmatrix} 8.02 & 8.20 & 8.00 & 8.14 & 7.92 \\ 7.98 & 8.10 & 8.00 & 7.90 & 8.06 \\ 7.92 & 8.04 & 8.04 & 7.84 & 7.94 \\ 8.08 & 8.06 & 8.02 & 7.82 & 8.04 \\ 8.14 & 8.00 & 7.98 & 7.96 & 8.02 \\ 7.86 & 7.94 & 8.18 & 8.02 & 7.92 \\ 7.96 & 7.98 & 7.96 & 8.04 & 8.00 \\ 8.06 & 8.12 & 7.94 & 7.98 & 7.98 \\ 7.88 & 7.92 & 8.16 & 8.02 & 8.00 \\ 7.90 & 8.02 & 8.04 & 8.02 & 8.02 \end{bmatrix}.''$$

It seems clear here that the individual is a resistor and the measurement X on an individual is the resistance in ohms. According to the authors, this collection of fifty resistors is the population. So according to Moore and McCabe (2003) if this is the population, then these are the only individuals in which the researcher is interested. Thus, resistors the production process may produce in the future are of no interest to the researcher. It is not clear the value of looking at any description of the variability found in the X values of these fifty resistors. On the other hand, we see that there is a process that is producing these resistors. One could view these fifty (actual) resistors as a sample that is representative of what the process can produce. It seems clear that there is potential for this process to produce more resistors with each having a resistance X. At this point, each of these potential resistors are conceptual with each being one of an uncountable collection of possibilities. Including with this uncountable collection of possibilities the fifty resistors the process has produced results in an uncountable collection which we have referred to as the population associated with this process. It seems that the quality practitioner would be interested in knowing, for example, what the average resistance μ is for the resistors being produced by this process. Note that μ is a parameter since it is describing the population. Since it is not possible to examine every resistor in the uncountable collection of resistors, then the practitioner must resort to using the sample data to obtain an estimate of μ. As we will see in the next chapter, the estimate for a parameter will typically be chosen to characterize the sample in a similar way that the parameter characterizes the population.

It is our belief that Kempthorne and Folks (1971) while using the word "population" to describe different collections of individuals a process has produced is usually using the word as a label of what we have referred to as the representative sample. What is not clear is how they would have labelled the next fifty resistors produced by the process. Would they refer to this collection as another population or would they combine the two collections and refer to this larger collection as the "population?"

We also find in more advance text, such as the mathematical statistics book by Bain and Engelhardt (1987) in which the authors on page 164 state "Perhaps it should be noted that definition 4.6.1 of a random sample does not apply to the case of 'random sampling without replacement' from a finite population." They go on to state "The random sample terminology also may refer here to the idea that each subset of size n elements of the populatin is equally likely to be selected as the sample. Definition 4.6.1 is applicable to sampling from finite populatins if the sampling is with replacement (or it would be approximately suitable if the population is quite large)." We do not concur with their statement about random sampling without replacement.

Box, Hunter, and Hunter (1978) on page 27 state "The total aggregate of observations that conceptually *might* occur as the result of performing a particular operation in a particular way is referred to as the *population* of observations. It is sometimes convenient to think of this population as infinite, but in this book we shall often think of it as finite and having size N, where N is large. The (usually few) observations that have actually occurred are thought of as some kind of *sample* from this population." On page 25, the authors state "An experimental result or datum is usually a numerical measurement that describes the outcome of the experimental run. For example, in the chemical experiment the datum of interest might be the yield of product. This would typically be defined as the amount of desired product formed, expressed as a percentage of the maximum theoretically attainable. Thus 10 successive runs made at what were intended to be identical conditions might give the following *data* in the form of percentage yields: 66.7, 64.3, 67.1, 66.1, 65.5, 69.1, 67.2, 68.1, 65.7, 66.4. Equally in the psychological experiment these data could be the times taken by 10 stressed subjects to perform a specific task." So for each of the aforementioned examples, who are the individuals? In the first it is what we will refer to as the product of a run of chemical experiment and in the second the individual is a stressed person. For each example, what is the population? Let's assume that in the first example, these are the only 10 runs that have been made. Are these 10 runs the population? Or does the population include the runs that "conceptually *might* occur?" In the second example, suppose it is the researcher who subjects the individuals to stress and each time the stress placed on an individual are not exactly identical (this is a reasonable assumption). Then these 10 individuals are the only actual individuals placed under the varying levels of stress from the population of interest. Is this set of 10 individuals the population? Or does the population consist of all the individuals under all possible levels of stress? If so, then the population is an uncountable collection of individuals the process has or could have generated. This results is a contradiction to Box, Hunter, and Hunter (1978) belief that the population is a finite collection.

George Edward Pelham Box

Wackerly, Mendenhall, and Scheaffer (2008) state on page 77 "As our final topic in this chapter, we move from theory to application and examine the nature of experiments conducted in statistics. A statistical experiment involves the observation of a sample selected from a larger body of data, existing or conceptual, called a *population*. The measurements in the sample, viewed as observations of the values of one or more random variables, are then employed to make an inference about the characteristics of the target population." Since one could argue that the collection of conceptual individuals is an uncountable collection of possibilities and the collection of existing individuals is a finite collection, then the population containing both is an uncountable collection. However, later in this same section, they state "suppose a fictitious population contains $N = 5$ elements, from which we plan to take a sample of size $n = 2$." This "fictitious" example suggest that they allow for the possibility of finite populations.

4. Graphical Methods for Displaying Sample Data

EXAMPLE 8. *A manufacturer of fertilizers selects sixteen farmers to use the company's new fertilizer for soybeans. It is believed that the new fertilizer will increase the yield of soybeans. Each of the farmers was asked to apply this fertilizer to a twenty acre plot. The farmers reported the following yields in bushels per acre: 35, 27, 43, 32, 31, 36, 37, 29, 30, 41, 25, 30, 29, 35, 37, 28. (a) Characterize an individual in the population of interest. (b) Describe the process that generates an individual from the population. (c) What is the measurement of interest? (d) What is the sample? (e) Make a dot plot of these data. (f) Discuss the distribution that generated these data. (g) Estimate (sketch) the density associated with the distribution of yields of soybeans measured in bushels per acre.*

Solution: (a) An "individual" is a 20-acre plot coupled with a growing season. (b) The process consists of "Mother Nature" selects a growing season for the 20-acre plot on which the farmer applies the manufacturer's fertilizer, grows, and harvests the soybeans. (c) The measurement of interest is the yield of soybeans on the 20-acre plot in units of bushels per acre. (A bushel of soybeans weights 60 pounds.) (d) The sample is the collection of the $n = 16$ twenty acre plots. (e) The following is a dot plot of these data.

```
          o o         o   o
  o   o o o o o o     o o o     o   o
------------------------------------------
 25        30        35      40        45
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(f) There appears to be one of two senerios that one could argue generated these data. (1) The distribution of yields can be described by a symmetric density curve or one that is very slightly skewed in the positive direction. (2) The density curve describing the distribution of yields is bimodal which suggest their are two populations from with the plots were selected. (g) The reader provides his/her guess.

EXAMPLE 9. *The following are test scores on the first exam given to 30 students taking an elementary statistic course: 75, 47, 84, 63, 70, 79, 83, 93, 85, 52, 85, 84, 67, 84, 72, 58, 87, 62, 88, 77, 80, 75, 73, 71, 92, 69, 91, 98, 76, 92. We will view test score as a continuous variable. What is the population of interest? What is the process that generates individuals from this population? Make a stemplot of these data. Discuss the distribution that generated these data. Estimate (sketch) the density associated with the distribution of test scores.*

Solution: There is a process that has generated the 30 students in the sample who take the first exam in this elementary statistics course. These students as well as the students the process could have generated constitute the population of interest. The measurement of interest on each student is the student's score on the first exam. The sample is the 30 students. A stem plot is these data is

```
9 -+- 3 2 1 8 2
8 -+- 4 3 5 5 4 4 7 8 0
7 -+- 5 0 9 2 7 5 3 1 6
6 -+- 3 7 2 9
5 -+- 2 8
4 -+- 7
```

We leave it to the reader to provide a sketch of a density curve that is an estimate of the density curve describing the distribution of test scores.

A histogram is a graphical method that was designed to be the sample counterpart to the density curve. It is a collection of bars placed over intervals with common endpoints. The area of each bar is the proportion of the observations that has a measurement in the corresponding interval. Also like its population counterpart, the density curve, the bars are on or above the measurement axis and the total area of the bars is one.

One method for constructing a histogram requires that the midpoint of one interval is the sample minimum ($X_{1:n}$) and the maximum ($X_{n:n}$) is the midpoint of another interval. Specifying the number of intervals between these two intervals and requiring the intervals to be of equal length, completely determines the intervals to be used. A simple rule one can use to select the number of intervals m to be

$m = \lceil n/5 \rceil$, that is, m is selected to be the smallest integer greater than or equal to $n/5$.

Requiring $X_{1:n}$ (minimum sample values) and $X_{n:n}$ (maximum sample value) as midpoints of the left and right most intervals that contain sample values and having the interval of equal length, the *width* of each interval is given by

$$w = \frac{X_{n:n} - X_{1:n}}{m-1} = \frac{\text{maximum sample value } - \text{minimum sample value}}{\text{one less than the number of intervals}}$$

The $m+1$ endpoints of these intervals are given by $X_{1:n} - w/2$, $X_{1:n} + w/2$, $X_{1:n} + 3w/2, \ldots, X_{1:n} + (2i-1)\,w/2, \ldots, X_{1:n} + (2m-1)\,w/2$. One problem that arises is when a data value is equal to one of the $m-1$ interior endpoints. While there are various ways to resolve this problem, some of which are quite elaborate, we will use a simple rule. We count a data value that falls on one of the endpoints in the right most interval.

We can now express the frequency and relative frequency distributions of our data in the following tabular form.

$$\begin{bmatrix} \begin{array}{c} \text{Interval} \\ \text{Number} \end{array} & \text{Interval} & \text{Frequency} & \begin{array}{c} \text{Relative} \\ \text{Frequency} \end{array} \\ 1 & \left[X_{1:n} - \frac{1}{2}w, X_{1:n} + \frac{1}{2}w\right) & f_1 & f_1/n \\ 2 & \left[X_{1:n} + \frac{1}{2}w, X_{1:n} + \frac{3}{2}w\right) & f_2 & f_2/n \\ \vdots & \vdots & \vdots & \vdots \\ i & \left[X_{1:n} + \frac{2i-3}{2}w, X_{1:n} + \frac{2i-1}{2}w\right) & f_i & f_i/n \\ \vdots & \vdots & \vdots & \vdots \\ m & \left[X_{1:n} + \frac{2m-3}{2}w, X_{1:n} + \frac{2m-1}{2}w\right) & f_m & f_m/n \end{bmatrix}$$

The height h_i of the ith bar must satisfy the equation $h_i w = f_i/n$. Solving for h_i, we have $h_i = f_i/(nw)$.

EXAMPLE 10. *The Food and Nutrition Board of the National Academy of Sciences states the recommended daily allowance (RDA) of iron is 18 mg for adult females under the age of 51. The amounts of iron intake during a 24-hour period for a sample of 45 such females are as follows. Data are in milligrams.*

$$\begin{bmatrix} 15.0 & 18.1 & 14.4 & 14.6 & 10.9 & 18.1 & 18.2 & 18.3 & 15.0 \\ 16.0 & 12.6 & 16.6 & 20.7 & 19.8 & 11.6 & 12.8 & 15.6 & 11.0 \\ 15.3 & 9.4 & 19.5 & 18.3 & 14.5 & 16.6 & 11.5 & 16.4 & 12.5 \\ 14.6 & 11.9 & 12.5 & 18.6 & 13.1 & 12.1 & 10.7 & 17.3 & 12.4 \\ 17.0 & 6.3 & 16.8 & 12.5 & 16.3 & 14.7 & 12.7 & 16.3 & 11.5 \end{bmatrix}$$

What is the population of interest? What is the measurement of interest? Construct a histogram using these data. Estimate (sketch) the density curve associated with the distribution of iron intake in milligrams during a 24-hour period for women under the age of 51. What can be said about the distribution that generated these data. ***Solution:*** *The population of interest is the collection of all adult women under the age of 51 years that the process of producing humans has or could have produced. The measurement of interest X is the amount of iron intake measured in milligrams in a 24 hour period.*

The following method for constructing a (relative frequency) histogram will be used to construct ALL histograms in this course unless otherwise directed.

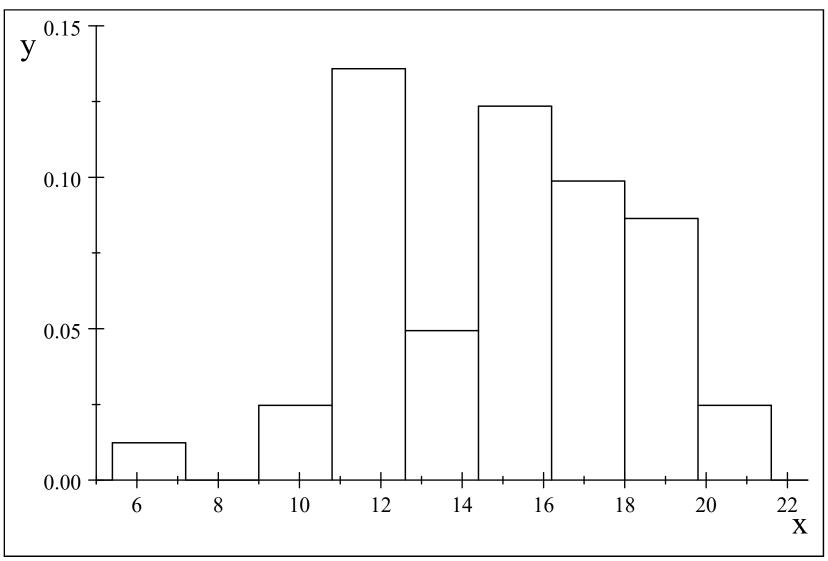
To construct the (relative frequency) histogram, we observe that $x_{45:1} = 6.3$ *milligrams and* $x_{45:45} = 20.7$ *milligrams. Using the rule* $m = \lceil n/5 \rceil$ *for selecting the number of intervals, then the number* m *of intervals would be selected to be* $m = 9$*. The width* w *of each interval is determined as*

$$w = (20.7 - 6.3) \,/\, (9 - 1) = 1.8$$

milligrams. The frequency and relative frequency distributions can now be determined and are given in the following table.

Interval Number	*Interval*	*Frequency*	*Relative Frequency*	*Bar Height*
1	$[5.4, 7.2)$	1	$\frac{1}{45}$	$\frac{1}{81}$
2	$[7.2, 9.0)$	0	$\frac{0}{45}$	$\frac{0}{81}$
3	$[9.0, 10.8)$	2	$\frac{2}{45}$	$\frac{2}{81}$
4	$[10.8, 12.6)$	11	$\frac{11}{45}$	$\frac{11}{81}$
5	$[12.6, 14.4)$	4	$\frac{4}{45}$	$\frac{4}{81}$
6	$[14.4, 16.2)$	10	$\frac{10}{45}$	$\frac{10}{81}$
7	$[16.2, 18.0)$	8	$\frac{8}{45}$	$\frac{8}{81}$
8	$[18.0, 19.8)$	7	$\frac{7}{45}$	$\frac{7}{81}$
9	$[19.8, 21.6)$	2	$\frac{2}{45}$	$\frac{2}{81}$

A (relative frequency) histogram can now be easily constructed. Complete the following graph of the (relative frequency) histogram for these data by drawing in the appropriate vertical line segments. Also, sketch your best estimate of the density curve describing the distribution of the amounts of iron intake in a 24-hour period for the population of adult women under the age of 51.



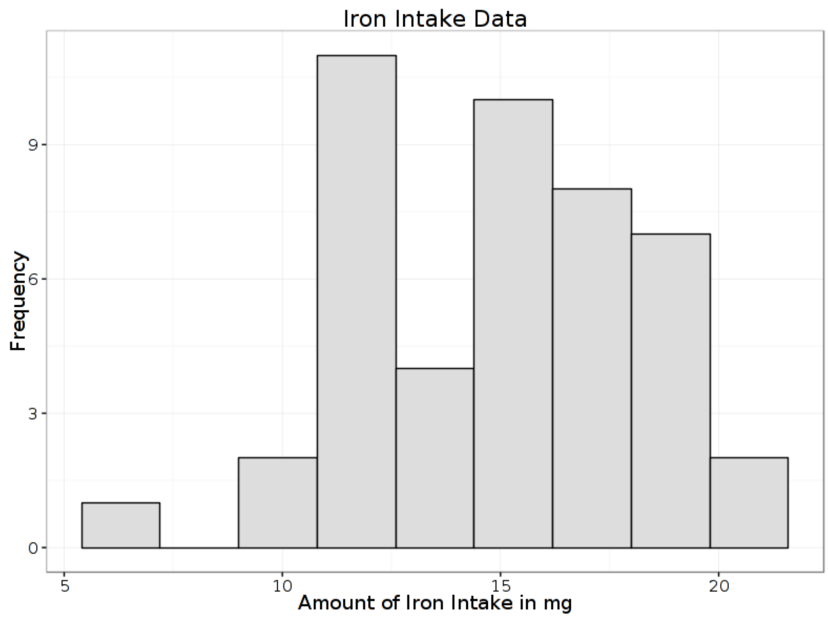


FIGURE 2

Using Crunchit!, we can obtain the frequency histogram presented in Figure 2.

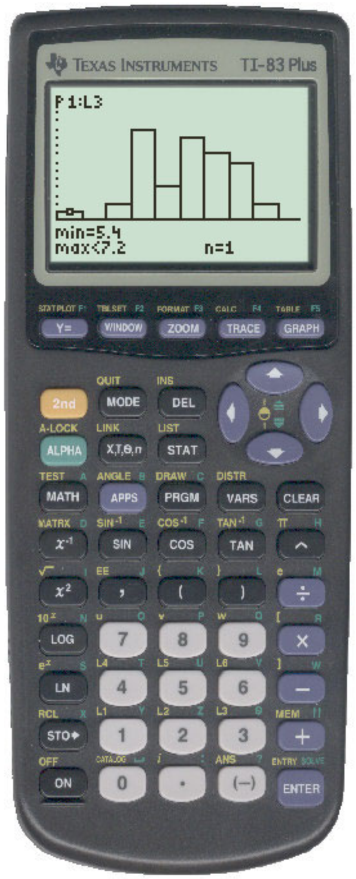


FIGURE 3

Using the TI-83 calculator, results in the same frequency histogram presented in the Figure 3. Note that the frequency histogram is proportional to the relative frequency histogram.

EXAMPLE 11. *Construct a (relative frequency) histogram and estimate (sketch) the density curve associated with this distribution of the test scores data: 75, 47, 84, 63, 70, 79, 83, 93, 85, 52, 85, 84, 67, 84, 72, 58, 87, 62, 88, 77, 80, 75, 73, 71, 92, 69, 91, 98, 76, 92.*

EXAMPLE 12. *From your knowledge of new car prices, estimate (sketch) the density curve associated with the distribution of new car prices. Is this distribution skewed in the positive or negative direction?* ***Solution:*** *Because there are new cars that are priced in the hundreds of thousands and even some in the millions dollars range, one would conclude that the distribution of new car prices are skewed in the positive direction. Any estimate of the density curve associated with this distribution should reflect this.*

EXAMPLE 13. *From your knowledge of new house prices, estimate (sketch) the density curve associated with the distribution of new home prices. Is this distribution skewed in the positive or negative direction?* ***Solution:*** *Because there are new houses that are priced in upper hundreds of thousands into the millions dollars range, one would conclude that the distribution of new house prices are skewed in the positive direction. Any estimate of the density curve associated with this distribution should reflect this.*

EXAMPLE 14. *From your knowledge, estimate (sketch) the density curve associated with the remaining lifetimes of American men age 35. Who would be interested in knowing this distribution? Is this distribution skewed in the positive or negative direction?* ***Solution:*** *We would expected that a American man age 35 would live on average about forty more years with these men beginning to die only several years from now. This suggest that the distribution of remaining lifetimes of American men age 35 would be skewed in the negative direction. Those companies furnishing life insurance policies would be interested in knowing this distribution (and those for other age groups) in order to determine the premium for the insurance policy.*

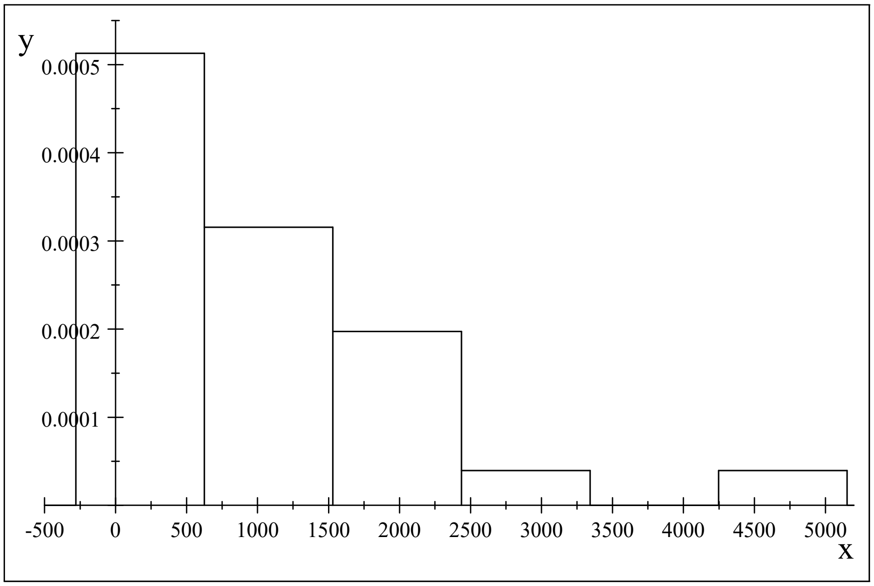
EXAMPLE 15. *A time plot is simply a plot of the measurement of interest verses a time variable. One example of a time plot is the plot of a stock's trading price verses time. Another is the plot of a quality statistic taken on the output of a production process verses the sample number. These time plots are referred to as quality control chart. They have been used successfully by practitioners in the production of quality goods. Control charts are discussed later in this chapter. Here is an example of the use of a time plot to examine the information found in data collected over time. In Exercise 1.28, page34 in Moore and McCabe (2006), "Recruitment," the addition of new members to a fish population, is an important measure of the health of ocean ecosystems. Here are data on the recruitment of rock* *sole in the Bering Sea between 1973 and 2000:*

Year	*Recruitment (millions)*	*Year*	*Recruitment (millions)*	*Year*	*Recruitment (millions)*
1973	173	1983	2246	1993	998
1974	234	1984	1793	1994	505
1975	616	1985	1793	1995	304
1976	344	1986	2809	1996	425
1977	515	1987	4700	1997	214
1978	576	1988	1702	1998	385
1979	727	1989	1119	1999	445
1980	1411	1990	2407	2000	676
1981	1431	1991	1049		
1982	1250	1992	505		

Make a graph to display the distribution of rock sole recruitment, then describe the pattern and any striking deviations that you see. ***Solution****: To examine the distribution of the variable "Recruitment," we will construct a histogram. The number of intervals is* $m = \lceil 28/5 \rceil = 6$ *with each interval having width* $w = (4700 - 173)\,/\,(6-1) = 905.4$*. The frequency and relative frequency distributions can now be determined and are given in the following table.*

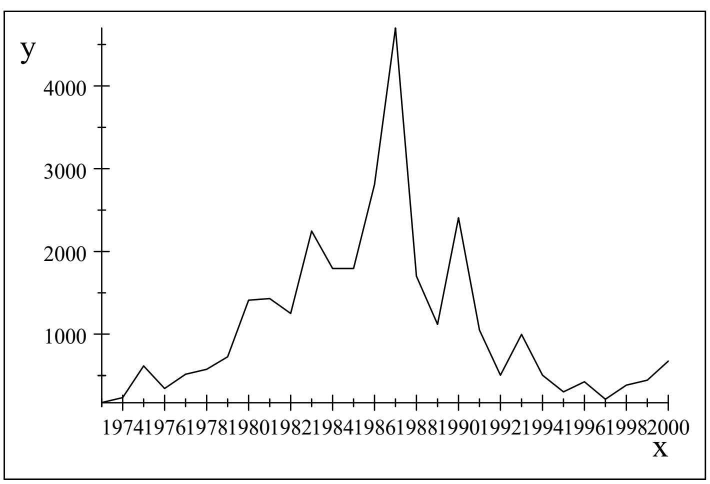
Interval Number	*Interval*	*Frequency*	*Relative Frequency*	*Bar Height*
1	$[-279.7, 625.7)$	13	$\frac{13}{28}$	$\frac{13}{25351.2}$
2	$[625.7, 1531.1)$	8	$\frac{8}{28}$	$\frac{8}{25351.2}$
3	$[1531.1, 2436.5)$	5	$\frac{5}{28}$	$\frac{5}{25351.2}$
4	$[2436.5, 3341.9)$	1	$\frac{1}{28}$	$\frac{1}{25351.2}$
5	$[3341.9, 4247.3)$	0	$\frac{0}{28}$	$\frac{0}{25351.2}$
6	$[4247.3, 5152.7)$	1	$\frac{1}{28}$	$\frac{1}{25351.2}$

The (relative frequency) histogram is given as follows with the vertical axis in units of $1/25351.2$*:*



An examination of the histogram suggest that the distribution is heavily skewed in the positive direction.

EXAMPLE 16. *(continuation) Make a time plot of recruitment and describe its pattern. As is often the case with time series data, a time plot is needed to understand what is happening. The distribution of the variable rock sole recruitment is skewed in the positive direction.* ***Solution****: A time series plot of "recruitment" verses year is given as follows. It appears that periodically relative large amounts of recruitment occurs.*



EXAMPLE 17. *The body temperture* X_t *of an individual varies with time* t*. Hence, a plot of body temperature* X_t *versus time* t *would be a time plot. Who would have an interest in viewing this time plot if it were available?*

5. Conclusion

Graphical displays of the sample data are quit useful in understanding the distribution of the measurement of interest. These include, among others, the dot plot, the stem plot, the (relative frequency) histogram, and time plots. Other plots not discussed are the pie and bar charts. While we did not discuss graphical representation of quality data in the form of a control chart, this is a type of time plot. Walter A. Shewhart introduced this graphical method in the early 1920's.

Walter A. Shewhart

Today, significant contributions in the area of quality control charting procedures including their application in biosurveillance has been made by Dr. Douglas C. Montgomery of Arizona State University and Dr. William H. Woodall of Virginia Tech.

Dr. William H. Woodall, Professor
Department of Statistics, Virginia Tech

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